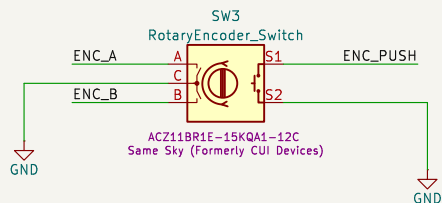
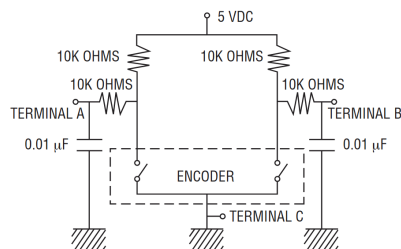


EC11 Encoder



For proper filtering
<https://www.bourns.com/docs/Product-Datasheets/PEC11R.pdf>
 Bourn has a recommended filtering schematic shown here:

Suggested Filter Circuit



Debounce

Dev Note: This breakout debounce approach is relying on the internal pullups of the microcontroller.
 In this case, I am assuming it's an nrf52, where according to a forum discussion the internal pullups is around 14kOhms.

I chose 100nF because it's a cheap capacitance value.

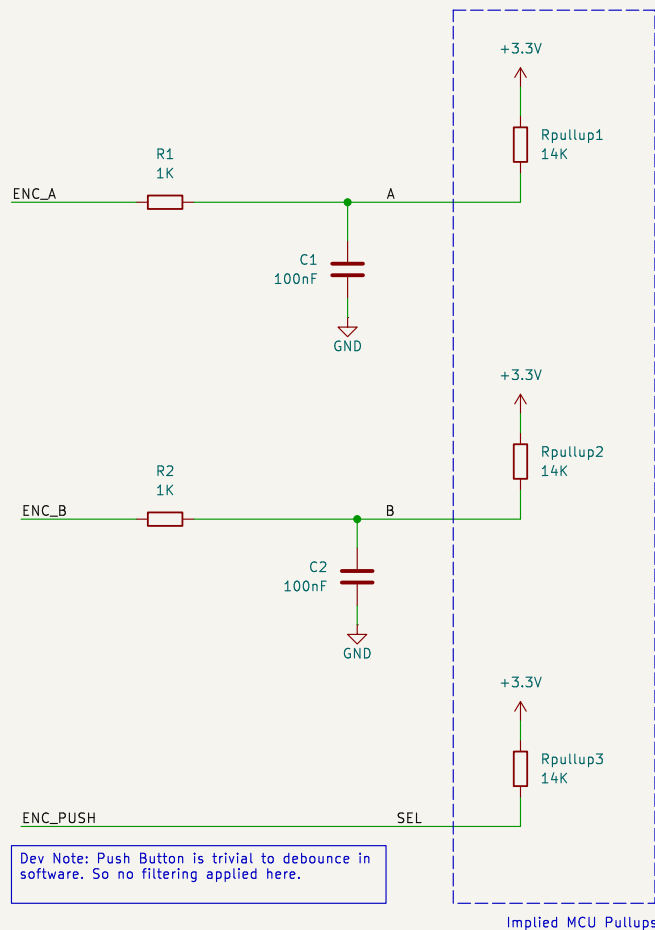
devzone.nordicsemi.com/f/nordic-q-a/107277/nrf52832-pull-up-pull-down-resistance

Figuring out switch resistance that would be below the 0.98V logic low level

$$\begin{aligned} V_s &= 3.3 \\ R_{pullup} &= 14k\Omega \\ R_{switch} &= 1k\Omega \\ V_{gpio} &= (V_s * R_{switch}) / (R_{pullup} + R_{switch}) \\ &= 0.22 \end{aligned}$$

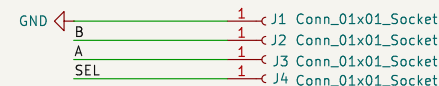
nRF52840 Product Specification v1.11
 6.9.3.1 GPIO Electrical Specification

$$\begin{aligned} \text{Input High Voltage Low Mark:} \\ &= 0.7 * V_{DD} \\ &= 0.7 * 3.3 \approx 2.31 \\ \text{Input Low Voltage High Mark:} \\ &= 0.3 * V_{DD} \\ &= 0.3 * 3.3 \approx 0.98 \end{aligned}$$

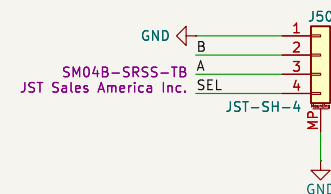


Dev Note: Push Button is trivial to debounce in software. So no filtering applied here.

Socket



Dev Note: This allows for point to point wire soldering



Dev Note: This is using the same connector as a common i2c qwiic connector.

Intended for use with https://github.com/mofosyne/roTec_pcb

Sheet: /
 File: ec11_horizontal_breakout_board.kicad_sch

Title: EC11 Horizontal Rotary Encoder Breakout Board

Size: A4 Date: 2025-12-27

KiCad E.D.A. 9.0.6

Rev: 3
 Id: 1/1